Testing pure components

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete pure-component test record.

Keep inputs, expected/actual outcomes, contract checks, detected defects and limits together.

Keep one habit: Recall → W1–W3/W5 → independent capacity9 → project unit cases → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

- Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

Least privilege and tool scopes

> **My remembered idea, example and correction**

Configuration and secrets

> **My remembered idea, example and correction**

Requirements and service boundaries

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- remaining(capacity, used) takes exact integers excluding bool with 0≤used≤capacity and returns capacity–used; invalid types or ranges raise ValueError. Supplied main cases (12, 5)→7; (12, 12)→0; (12, 13)→ValueError rather than–1. Zero-capacity/zero-used is valid. A pure component depends on supplied inputs and has no external effects; no clock/network/model/booking is needed.

  Separate picture formatter takes a positive integer and returns exact "Room " plus its digits:7→"Room 7", 12→"Room 12"; zero rejects. Do not copy its positive-only boundary into the remaining-count contract. Exact expected outcomes are derived before running. The harness records values versus errors, checks a finite invariant range and detects an intentionally faulty extra subtraction.

  Unit evidence does not establish integration permissions, real room records, model behaviour or every possible input.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- remaining(capacity, used) takes exact integers excluding bool with 0≤used≤capacity and returns capacity–used; invalid types or ranges raise ValueError. Supplied main cases (12, 5)→7; (12, 12)→0; (12, 13)→ValueError rather than–1. Zero-capacity/zero-used is valid. A pure component depends on supplied inputs and has no external effects; no clock/network/model/booking is needed.

  Separate picture formatter takes a positive integer and returns exact "Room " plus its digits:7→"Room 7", 12→"Room 12"; zero rejects. Do not copy its positive-only boundary into the remaining-count contract. Exact expected outcomes are derived before running. The harness records values versus errors, checks a finite invariant range and detects an intentionally faulty extra subtraction.

  Unit evidence does not establish integration permissions, real room records, model behaviour or every possible input.

- Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- remaining(capacity, used) takes exact integers excluding bool with 0≤used≤capacity and returns capacity–used; invalid types or ranges raise ValueError. Supplied main cases (12, 5)→7; (12, 12)→0; (12, 13)→ValueError rather than–1. Zero-capacity/zero-used is valid. A pure component depends on supplied inputs and has no external effects; no clock/network/model/booking is needed.

  Separate picture formatter takes a positive integer and returns exact "Room " plus its digits:7→"Room 7", 12→"Room 12"; zero rejects. Do not copy its positive-only boundary into the remaining-count contract. Exact expected outcomes are derived before running. The harness records values versus errors, checks a finite invariant range and detects an intentionally faulty extra subtraction.

  Unit evidence does not establish integration permissions, real room records, model behaviour or every possible input.

- A report omits this qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- remaining(capacity, used) takes exact integers excluding bool with 0≤used≤capacity and returns capacity–used; invalid types or ranges raise ValueError. Supplied main cases (12, 5)→7; (12, 12)→0; (12, 13)→ValueError rather than–1. Zero-capacity/zero-used is valid. A pure component depends on supplied inputs and has no external effects; no clock/network/model/booking is needed.

  Separate picture formatter takes a positive integer and returns exact "Room " plus its digits:7→"Room 7", 12→"Room 12"; zero rejects. Do not copy its positive-only boundary into the remaining-count contract. Exact expected outcomes are derived before running. The harness records values versus errors, checks a finite invariant range and detects an intentionally faulty extra subtraction.

  Unit evidence does not establish integration permissions, real room records, model behaviour or every possible input.

- For the worked scenario, propose one additional case that could reveal a failure, describe its **expected outcome** before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled testing pure components **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing pure components.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Testing pure components. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Testing pure components. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

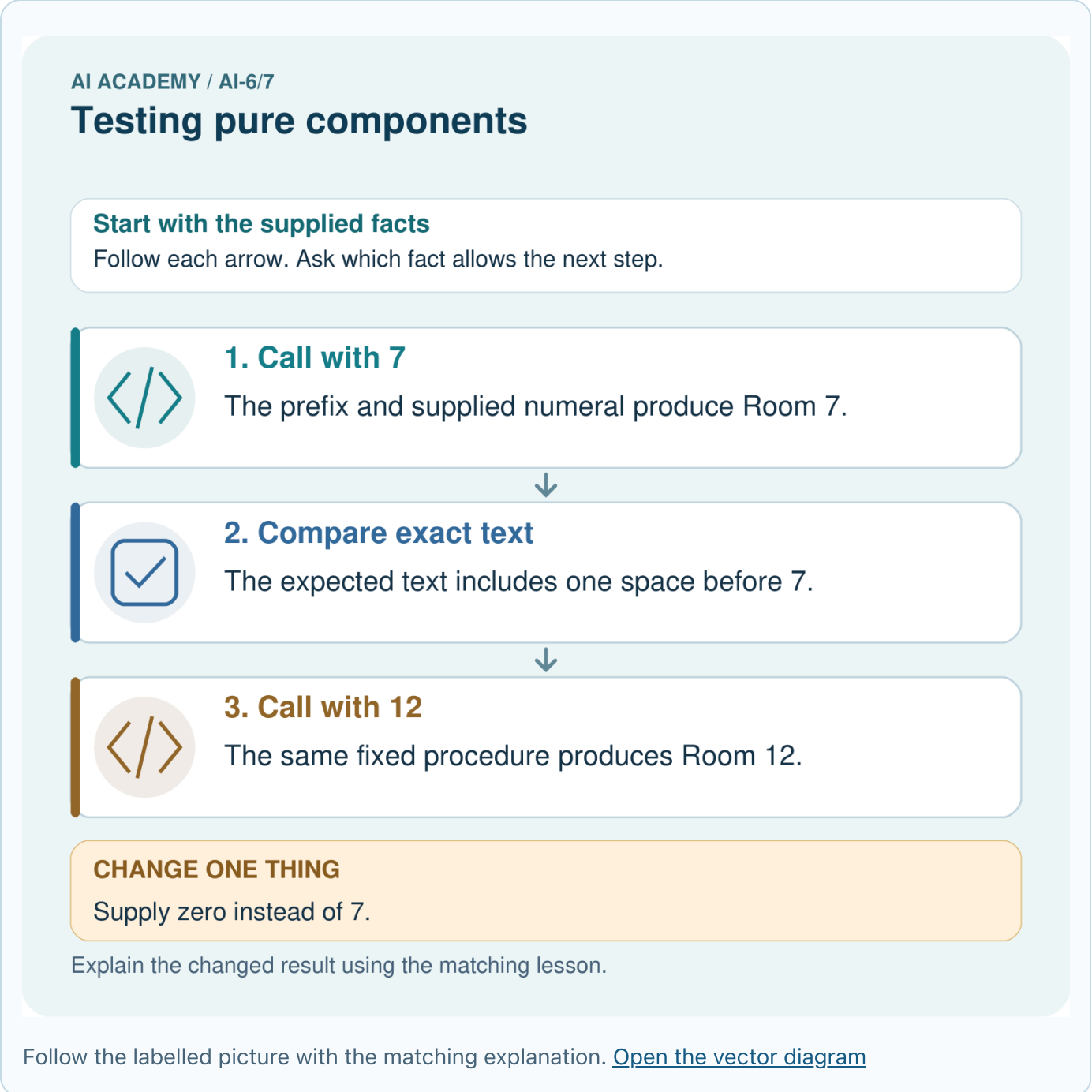
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.
- Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".
- Predict the formatter **output**s for 7 and 12, including spacing. Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.
- Which labelled step matches this detail? The same fixed procedure produces Room 12.
- 1. Call with 12
- 2. Compare exact text
- 3. Call with 7

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Call with 12 The same fixed procedure produces Room 12. A pure component depends on its accepted **input**, not a clock, network or model.

Isolating it makes failures easier to locate.

Changed-case question and optional hint

- Changed-case practice: Supply zero instead of 7. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Reject the **input** under the stated positive-integer contract.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Reject the **input** under the stated positive-integer contract. Zero has the right type but violates the value requirement.

Repair and retain the record

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor’s request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

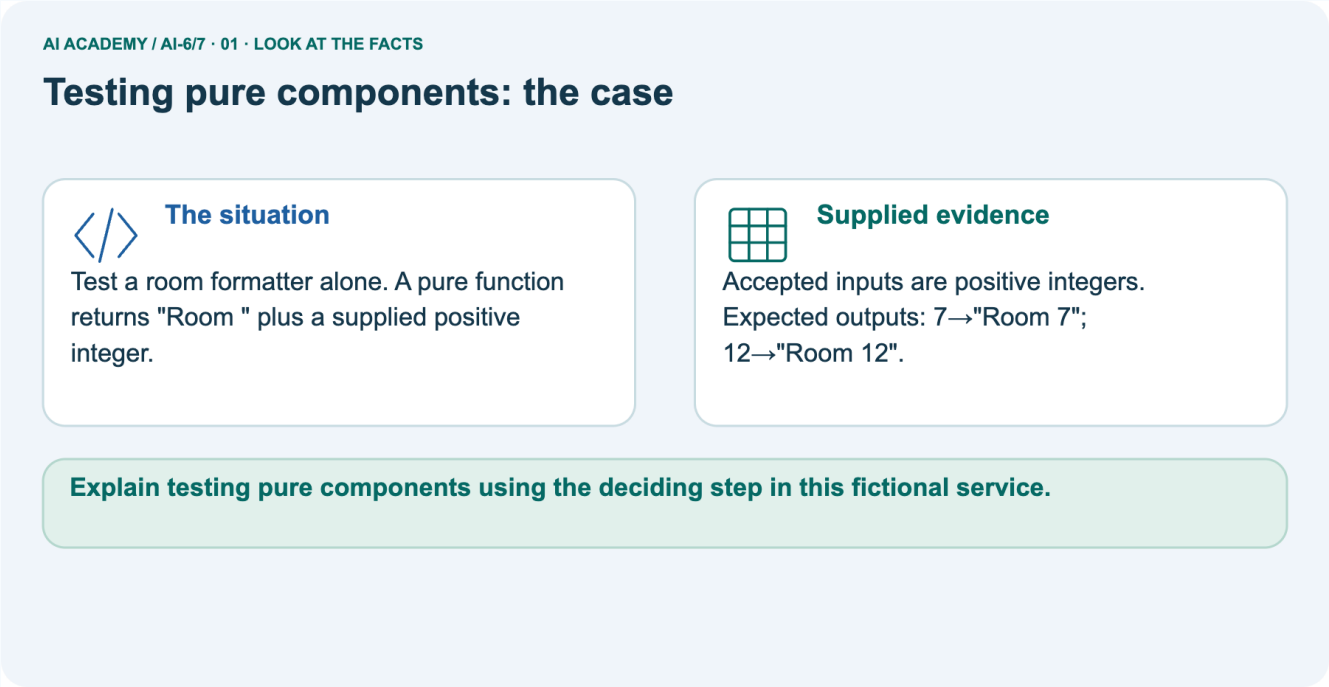
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Testing pure components: the case



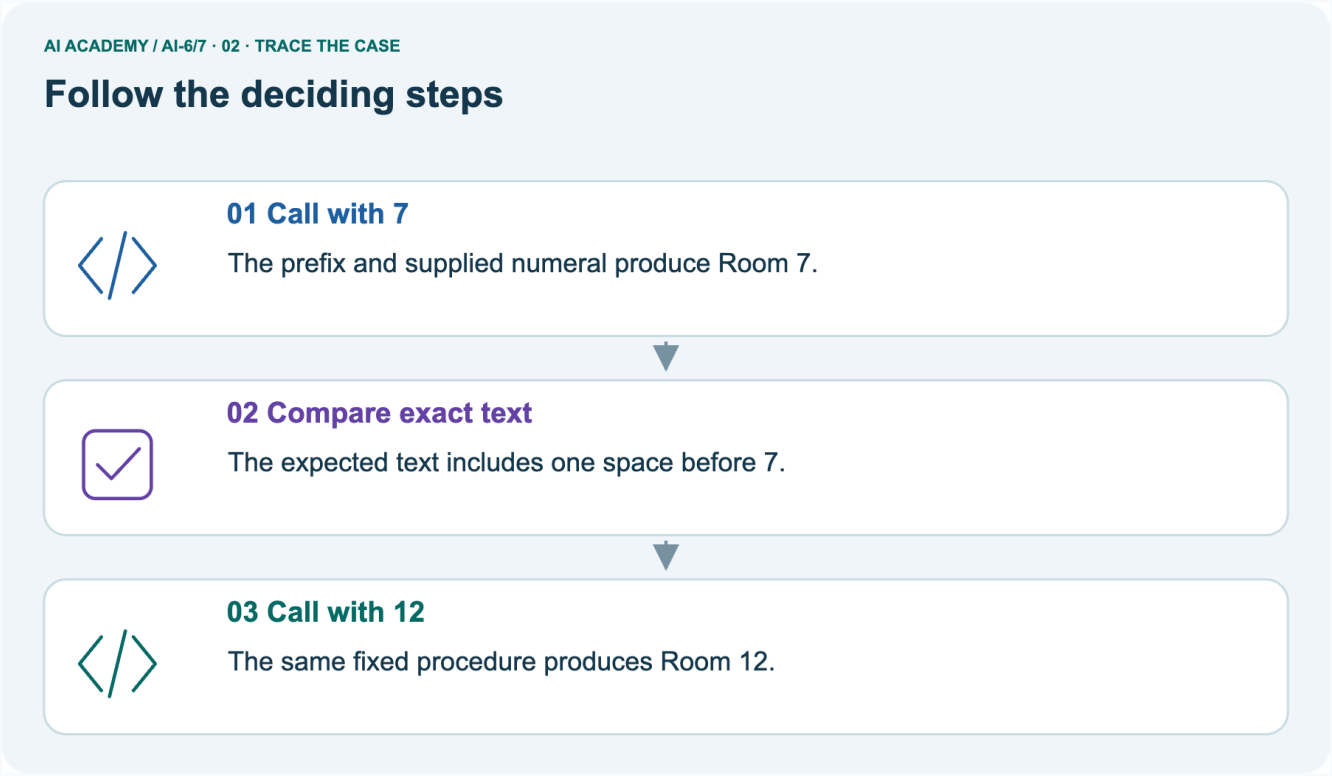
Follow the labelled picture with the matching explanation. Open the vector diagram

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer. Supplied facts: Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



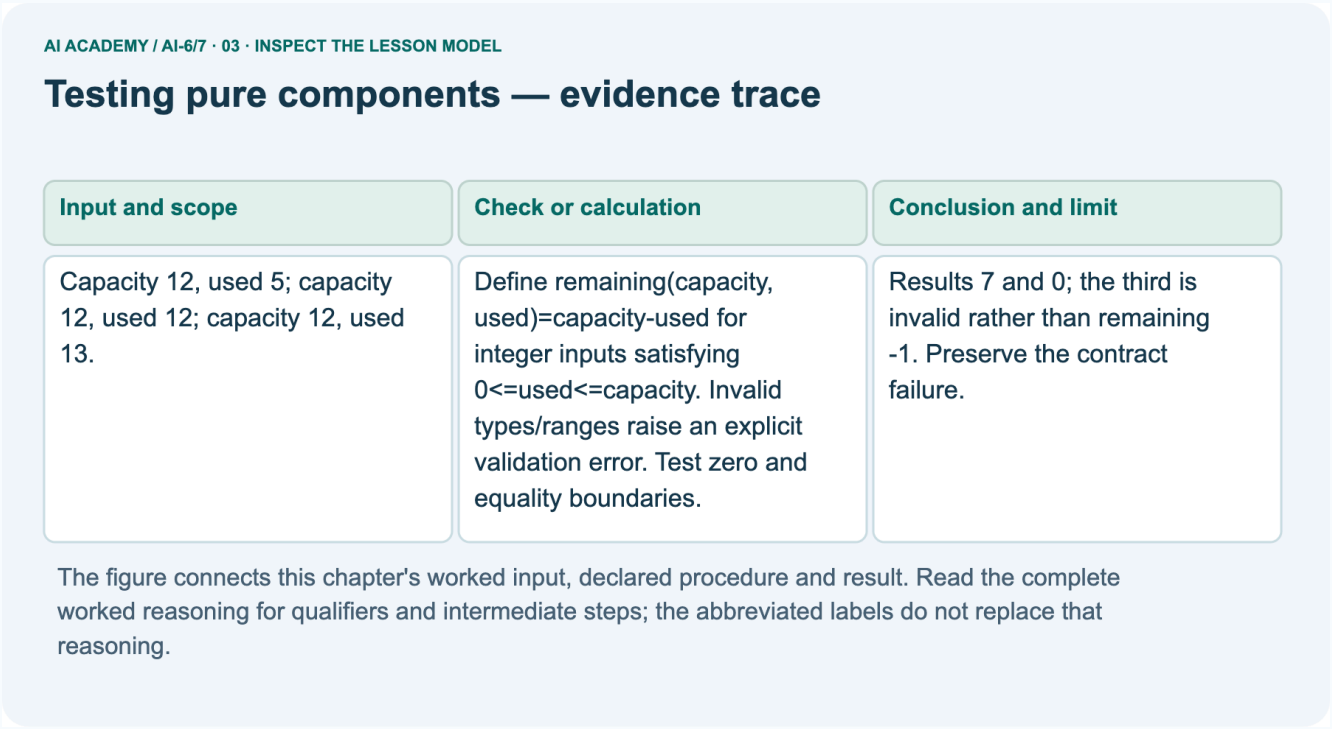
Follow the labelled picture with the matching explanation. Open the vector diagram

Call with 7: The prefix and supplied numeral produce Room 7. → Compare exact text: The expected text includes one space before 7. → Call with 12: The same fixed procedure produces Room 12.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Testing pure components — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

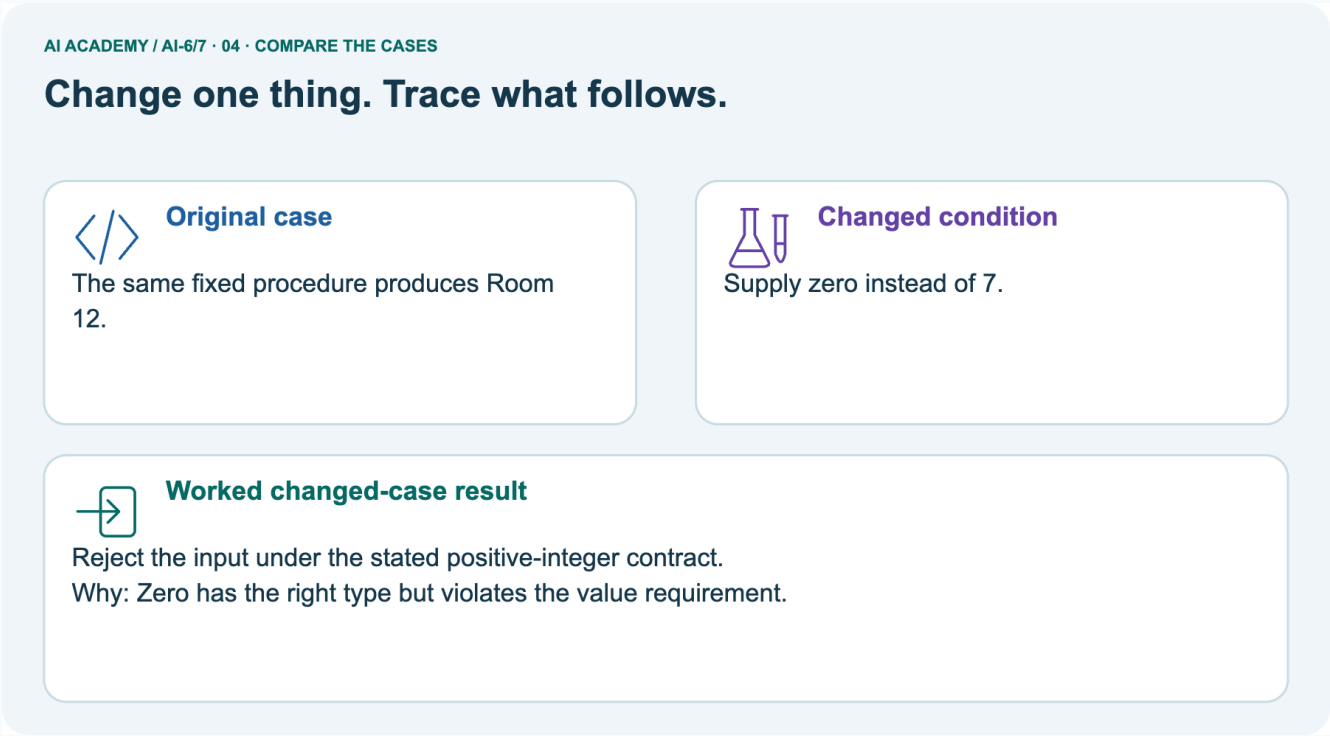
Testing pure components — **evidence** trace. Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. | Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity.

Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries. | Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

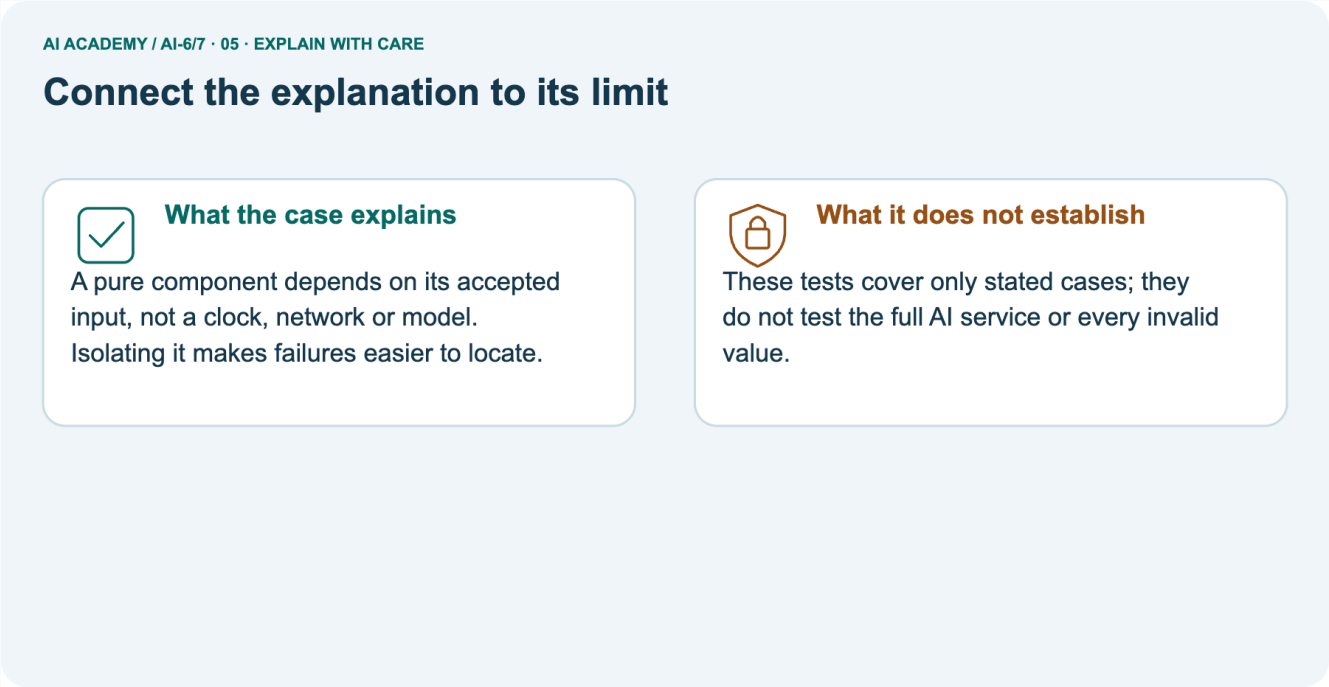
Original case: The same fixed procedure produces Room 12. Changed condition: Supply zero instead of 7. Expected result: Reject the **input** under the stated positive-integer contract.

Reason: Zero has the right type but violates the value requirement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

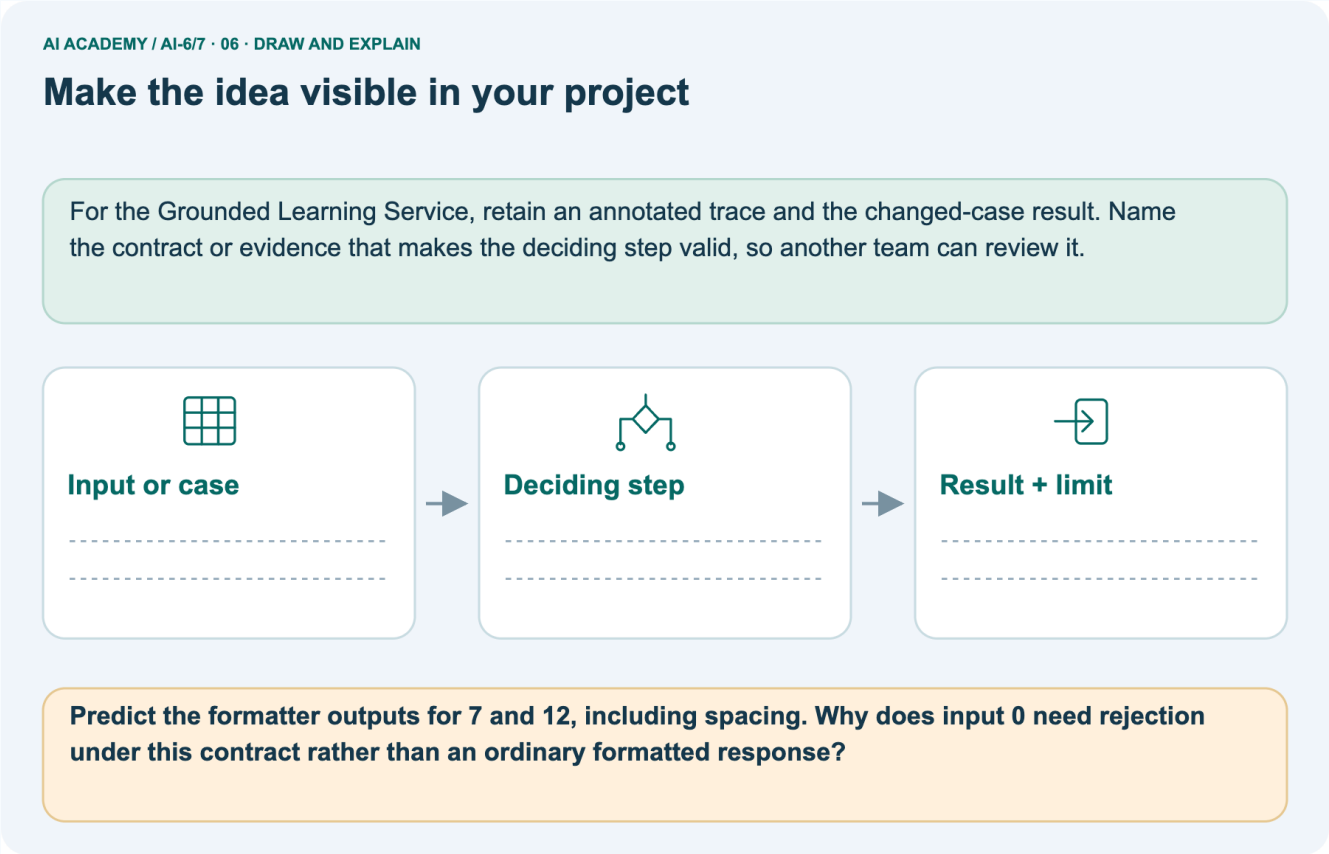
A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

Limit: These tests cover only stated cases; they do not test the full AI service or every invalid value.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Testing pure components: the case

02. Follow the deciding steps

03. Testing pure components — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.